

# **MTU\_ValueService** **Technical Documentation**

**Exhaust Flap Control Unit**  
**FCU 2-01**  
**FCU 1-01**  
**for Blue Line Series 4000**  
**Application: Marine**

Functional Description  
Operating Instructions  
Workshop Manual  
Installation and Commissioning Instructions

**E532215/01E**



Printed in Germany

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This handbook is provided for use by maintenance and operating personnel in order to avoid malfunctions or damage during operation.

Subject to alterations and amendments.

# Commissioning Note

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## Important

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Please complete and return the “Commissioning Note” card below to MTU Friedrichshafen GmbH.

The Commissioning Note information serves as a basis for the contractually agreed logistic support (warranty, spare parts, etc.).



Postcard  
MTU Friedrichshafen GmbH  
Technical Information Management  
Dept. VOT  
88040 Friedrichshafen  
GERMANY

# Commissioning Note

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*Please use block capitals!*



|                                                                                                                                                    |                          |
|----------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| Engine No.:                                                                                                                                        | MTU works order no.:     |
| Engine model:                                                                                                                                      | Date put into operation: |
| Application : *<br><input type="checkbox"/> Marine <input type="checkbox"/> Rail<br><input type="checkbox"/> Genset <input type="checkbox"/> ..... | Type :<br>Manufacturer:  |
| End user`s address:                                                                                                                                |                          |
| Remarks:                                                                                                                                           |                          |

**Commissioning  
Note**

**\***  
Tick appropriate box

|          |                                                     |           |
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# 1 General Provisions

## 1.1 Safety

### 1.1.1 General conditions

#### General

In addition to the instructions in this publication, the applicable country-specific legislation and other compulsory regulations regarding accident prevention and environmental protection must be observed. This state-of-the-art engine has been designed to meet all applicable laws and regulations. The engine may nevertheless present a risk of injury or damage in the following cases:

- Incorrect use
- Operation, maintenance and repair by unqualified personnel
- Modifications or conversions
- Noncompliance with the Safety Instructions

#### Correct use

The engine is intended solely for use in accordance with contractual agreements and the purpose envisaged for it on delivery. Any other use is considered improper use. The engine manufacturer accepts no liability whatsoever for resultant damage or injury in such case. The responsibility is borne by the user alone.

Correct use also includes observation of and compliance with the maintenance specifications.

#### Modifications or conversions

Unauthorized modifications to the engine represent a safety risk.

MTU will accept no liability or warranty claims for any damage caused by unauthorized modifications or conversions.

#### Spare parts

Only genuine MTU spare parts must be used to replace components or assemblies. MTU accepts no liability whatsoever for damage or injury resulting from the use of other spare parts and the warranty shall be voided in such case.

#### Reworking components

Repair or engine overhaul must be carried out in workshops authorized by MTU.

## 1.1.2 Personnel and organizational requirements

### Personnel requirements

All work on the engine shall be carried out by trained and qualified personnel only.

The specified legal minimum age must be observed.

The operator must specify the responsibilities of the operating, maintenance and repair personnel.

### Organizational measures

This publication must be issued to all personnel involved in operation, maintenance, repair or transportation.

Keep it at hand at the operating site of the engine so that it is available to operating, maintenance, repair and transport personnel at all times.

Use the manual as a basis for instructing personnel on engine operation and repair with an emphasis on explaining safety-relevant instructions.

This is particularly important in the case of personnel who only occasionally perform work on or around the engine.

This personnel must be instructed repeatedly.

For the identification and layout of the spare parts during maintenance or repair work, take photos or use the spare parts catalog.

### Working clothes and protective equipment

Wear proper protective clothing for all work.

Use the necessary protective equipment for the given work to be done.



### 1.1.3 Safety regulations for maintenance and repair work

#### Safety regulations for maintenance and repair work

- Have maintenance and repair work carried out by qualified and authorized personnel only.
- Allow the engine to cool down before starting maintenance work (risk of explosion of oil vapors).
- Before starting work, relieve pressure in systems and compressed-air lines which are to be opened.
- Take special care when removing ventilation or plug screws from the engine. Cover the screw or plug with a rag to prevent fluids escaping under pressure.
- Take special care when draining hot fluids ⇒ Risk of injury.
- When changing the engine oil or working on the fuel system, ensure that the engine room is adequately ventilated.
- Allow the engine / system to cool down before starting to work.
- Observe the maintenance and repair instructions.
- Never carry out maintenance and repair work with the engine running unless expressly instructed to do so.
- Secure the engine against accidental starting.
- Disconnect the battery when electrical starters are fitted.
- Close the main valve on the compressed-air system and vent the compressed-air line when pneumatic starters are fitted.
- Disconnect the control equipment from the assembly or system.
- Use only proper, calibrated tools. Observe the specified tightening torques during assembly/disassembly.
- Carry out work only on assemblies and/or units which are properly secured.
- Never use lines for climbing.
- Keep fuel injection lines and connections clean.
- Always seal connections with caps or covers if a line is removed or opened.
- Take care not to damage lines, in particular fuel lines, during maintenance and repair work.
- Ensure that all retainers and dampers are installed correctly.
- Ensure that all fuel injection and pressurized oil lines are installed with enough clearance to prevent contact with other components. Do not place fuel or oil lines near hot components.
- Do not touch elastomeric seals if they have carbonized or resinous appearance unless hands are properly protected.
- Note cooling time for components which are heated for installation or removal ⇒ Risk of burning.
- When working high on the engine, always use suitable ladders and work platforms. Make sure components are placed on stable surfaces.
- Observe special cleanliness when conducting maintenance and repair work on the assembly or system. After completion of maintenance and repair work, make sure that no loose objects are in/on the assembly or system.
- Before barring the engine, make sure that nobody is standing in the danger zone. Check that all guards have been reinstalled and that all tools and loose parts have been removed after working on the engine.
- The following additional instructions apply to starters with beryllium copper pinion:  
Breathing protection of filter class P2 must be applied during maintenance work to avoid health hazards caused by the beryllium-containing pinion. Do not blow out the interior of the flywheel housing or the starter with compressed air. Clean the flywheel housing inside with a class H dust extraction device as an additional measure.

#### Welding work

Never carry out welding work on the assembly, system, or engine-mounted units. Cover the engine when welding in its vicinity.

Do not use the assembly or system as ground terminal.

Do not route the welding lead over or near the wiring harnesses of MTU systems. The welding current may otherwise induce an interference voltage in the wiring harnesses which could conceivably damage the electrical system.

Remove parts (e.g. exhaust pipes) which are to be welded from the engine beforehand.

#### Hydraulic installation and removal

Check the function and safe operating condition of tools and fixtures to be used. Use only the specified fixtures for hydraulic removal/installation procedures.

Observe the max. permissible push-on pressure specified for the equipment.

Do not attempt to bend or apply force to lines.

Before starting work, pay attention to the following:

- Bleed air from jigs and fixtures for force-fitting/removing parts, pumps and piping at the points provided for the system concerned (e.g. open threaded vent plugs, pump until no bubbles can be discerned in the oil, close threaded vent plugs).
- For hydraulic installation, screw on the tool with the piston retracted.
- For hydraulic removal, screw on the tool with the piston extended.

For a hydraulic installation/removal tool with central expansion pressure supply, screw spindle into shaft end until correct sealing is established.

During hydraulic installation and removal, ensure that nobody is standing in the immediate vicinity of the component to be installed/removed.

### **Working on electrical/electronic assemblies**

Always obtain the permission of the person in charge before commencing maintenance and repair work or switching off any part of the electronic system required to do so.

De-energize the appropriate areas prior to working on assemblies.

Do not damage cabling during removal work. When reinstalling ensure that cabling is not damaged during operation by contact with sharp objects, by rubbing against other components or by a hot surface.

Do not secure cables on lines carrying fluids.

Do not use cable binders to secure cables.

Always use connector pliers to tighten connectors.

Subject the device or system to a function check on completion of all repair work.

Store spare parts properly prior to replacement, i.e. protect them against moisture in particular. Pack defective electronic components and assemblies in a suitable manner when dispatched for repair, i.e. particularly protected against moisture and impact and wrapped in antistatic foil if necessary.

### **Working with laser equipment**

When working with laser equipment, always wear special laser-protection goggles ⇒ Heavily focused radiation.

Laser equipment must be fitted with the protective devices necessary for safe operation according to type and application.

For conducting light-beam procedures and measurement work, only the following laser devices must be used:

- Laser devices of classes 1, 2 or 3A.
- Laser devices of class 3B, which have maximum output in the visible wavelength range (400 to 700 nm), a maximum output of 5 mW, and in which the beam axis and surface are designed to prevent any risk to the eyes.